

Graduation Project Proposal

Finaira Reversible Privacy Gateway for Arabic and English Financial Documents

Sponsored by Finaira

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1.0 Problem Statement and Background

Financial institutions increasingly rely on powerful cloud artificial intelligence models for document search, question answering, summarization, and information extraction. The same documents may contain customer names, account numbers, contact details, national identifiers, transaction references, and other confidential relationships that should not leave the institution in raw form. The Finaira project therefore investigates a local privacy gateway that transforms sensitive information before an employee request or document crosses the boundary to an external model, then restores authorized information after the response returns [1].

The required transformation is reversible. A value such as a customer name or account number is replaced locally with an opaque typed token, while the reverse mapping remains encrypted and controlled inside the organization. This process is pseudonymization rather than permanent anonymization because the original value can still be recovered using separately stored information. Text anonymization research also shows that direct identifier removal alone is insufficient: context and combinations of quasi-identifiers may still reveal a person, while aggressive masking can damage the usefulness of the text [2].

The general pattern of hiding sensitive text before cloud inference is established prior art. HaS performs local hiding and later de-anonymization around an external language model [3], while Portcullis demonstrates a scalable gateway based on substitution and secure reconstruction [4]. The proposed project will therefore not claim that it invented the privacy gateway. Its contribution will be the adaptation and evaluation of this pattern for Arabic and English financial documents, mixed-language text, ambiguous numbers, configurable policy, employee chat context, cross-document token consistency, and complete cloud round trips.

Arabic financial-named entity recognition provides useful starting points but does not solve the privacy problem. AMWAL reports strong aggregate results on Arabic financial news, yet its labels describe financial concepts and do not cover the full privacy taxonomy required for account numbers, IBANs, cards, telephone numbers, email addresses, and national identifiers [5]. Wojood provides a large Arabic named entity corpus that supports general entities such as people, organizations, and locations, but it is not a financial PII benchmark [6]. GLiNER2 PII offers a flexible 42-type PII schema and exact character spans, but its published training and evaluation do not establish performance on Arabic financial documents [7].

The project must consequently answer a system-level question rather than only a model-selection question. It must determine whether a hybrid of validated rules, Arabic and multilingual models, contextual resolution, policy-controlled pseudonymization, and deterministic safety gates can reduce disclosure while preserving retrieval and answer quality for authorized bank employees.

1.1 Research Question

How effectively can a local policy-driven privacy gateway detect and reversibly pseudonymize sensitive information in Arabic, English, and mixed-language financial documents while preserving the usefulness and integrity of cloud-based retrieval and question answering?

2.0 Project Goal and Objectives

2.1 Project Goal

The project goal is to design, implement, and evaluate an employee-facing privacy gateway that allows authorized financial staff to use cloud AI capabilities without sending detected raw sensitive information beyond the organization's trusted environment.

2.2 Project Objectives

1. Define a Finaira financial privacy taxonomy that separates entity type from the policy action applied to that entity.
2. Create a local document-preparation and normalization pipeline for Arabic, English, and mixed-language text while preserving document structure and character positions.
3. Benchmark deterministic rules and checksums, Arabic and multilingual entity models, and a selective local contextual model for privacy-sensitive span detection.
4. Implement a fusion and context-resolution layer that handles overlapping detections, repeated aliases, code-switching, and ambiguous financial numbers.
5. Implement a versioned privacy policy that can preserve, tokenize, generalize, remove, block, or request review for each entity type.
6. Build scoped reversible pseudonymization with an encrypted local mapping vault and exact authorized restoration.
7. Develop an employee-facing retrieval and question-answering prototype that sends only sanitized queries, conversation context, and document chunks to a cloud model.
8. Implement pre-egress and post-response safety gates for residual PII, unresolved risk, token integrity, and unexpected sensitive output.
9. Construct a human-reviewed benchmark and report per-type detection quality, document leakage, reconstruction, retrieval and answer utility, robustness, and latency.

3.0 Scope

The proposed system is an internal assistant for authorized bank employees. It is not a customer-facing chatbot. The prototype will demonstrate how an employee can search or ask questions about approved financial documents while the local gateway controls what reaches the cloud model.

Boundary	Description
Included	Arabic, English, and mixed-language financial text
Included	Plain text and text-layer PDF documents for the initial vertical slice
Included	Selected structured identifiers, people, organizations, locations, and other agreed sensitive categories
Included	Sanitized employee queries, session history, retrieved chunks, and cloud responses
Included	Configurable request, document, or corpus token scope
Included	One primary cloud task, proposed as retrieval question answering

The final document formats, supported jurisdictions and identifiers, local hardware, cloud provider, retention rules, and acceptable false-negative thresholds require confirmation from Finaira. These choices affect the recognizers, policies, validation rules, and evaluation datasets.

4.0 Proposed System Architecture

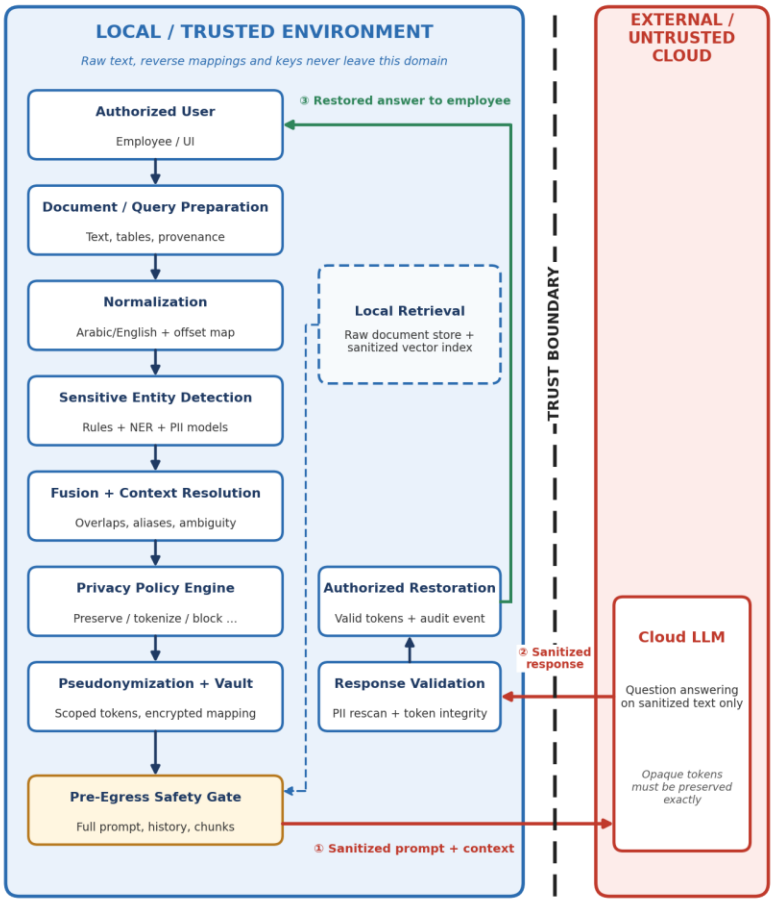


Figure 1 Proposed Finaira system architecture and trust boundary

The proposed system is organized around a local trusted gateway placed between the authorized employee and the external cloud model. Inside the trusted environment, submitted documents and queries are prepared and normalized, after which the detection stage combines deterministic rules, validators, and PII and named-entity models to propose sensitive spans. Fusion and context resolution convert these candidates into final entities, and the policy engine assigns each entity a preserve, tokenize, generalize, remove, review, or block action. Protected spans are replaced by scoped opaque tokens, and the reverse mapping is held in an encrypted local vault that never crosses the trust boundary; the raw document store and the retrieval index built from sanitized text likewise remain local. A pre-egress safety gate validates the fully assembled prompt, history, and retrieved chunks, and only then does sanitized content leave the trusted environment for the cloud model, which performs retrieval question answering using opaque tokens alone. The response re-enters the gateway, where validation rejects invented, malformed, or cross-session tokens and rescans for unexpected sensitive output, and authorized restoration then replaces valid tokens with original values only for employees entitled to view them. The individual stages are specified in Section 6.3.

5.0 Significance of the Project

The project addresses a practical barrier to using external AI services with confidential financial documents. A functioning gateway would demonstrate how raw identifiers can remain local while employees still benefit from cloud-based document understanding. The design also separates detection from policy, allowing the same technical recognizers to support different organizational rules without embedding policy decisions inside a model.

The academic value lies in evaluation rather than the existence of a proxy. Recent subject-level research shows that high span recall can coexist with substantial contextual inference risk [8]. Prior pseudonymization studies also show that transformations affect downstream tasks differently, so semantic similarity alone cannot demonstrate preserved utility [9]. Finaira can contribute a realistic Arabic and English financial benchmark, a transparent error analysis, and an end-to-end privacy-utility study that measures leakage, retrieval, question answering, token integrity, and exact restoration together.

The resulting prototype is also a useful engineering artifact. It combines document processing, Arabic language handling, deterministic validation, machine learning inference, secure storage, retrieval, cloud integration, access-aware restoration, testing, and privacy-safe observability in one demonstrable system.

5.1 Expected Contributions

The project builds on established work and does not claim it as its own. The general pattern of hiding sensitive text before cloud inference [3], [4], pseudonymization as a privacy technique [12], [13], and existing detection resources and tools such as Presidio, AMWAL, Wojood, and GLiNER2 PII [5]-[7], [16] are prior art, and will serve as foundations or baselines. The expected contributions of Finaira lie in the integration, specialization, evaluation, and benchmark work listed below. They are the anticipated outcomes of the proposed project and are not presented as completed results.

1. **Privacy gateway prototype.** An employee-facing gateway prototype that adapts the established local-gateway pattern to Arabic, English, and mixed-language financial documents, covering the complete round trip from local sanitization through cloud question answering to authorized restoration.
2. **Hybrid sensitive-information detection pipeline.** A pipeline that combines deterministic rules, financial identifier validators, Arabic and multilingual named-entity and PII models, and selective contextual resolution, together with a detector ablation that quantifies what each component contributes. No single model is treated as the safety boundary.
3. **Configurable privacy-policy framework.** A versioned policy layer that separates entity type from the action applied to it, so that the same recognizers can support different organizational rules without embedding policy decisions inside a model.

4. **Reversible pseudonymization mechanism.** Scoped opaque tokens with an encrypted local mapping vault and exact authorized restoration, accompanied by a token-scope comparison that measures the utility gained against the linkability introduced.
5. **Human-reviewed Arabic and English financial privacy benchmark.** A benchmark built from controlled financial templates, public resources, and human-reviewed Finaira-like examples, covering Arabic-only, English-only, and code-switched text, digit variants, transliteration, structured identifiers, hard negatives, and ambiguous financial numbers, with entity type, sensitivity, expected policy action, and co-reference annotated separately. Existing benchmarks such as TAB and REDACT [10], [11] will serve as external checks rather than substitutes.
6. **End-to-end privacy-utility evaluation.** A joint evaluation of detection quality, document leakage, retrieval, question answering, reconstruction, robustness, and system performance, intended to show privacy and utility together rather than span-level detection alone.

6.0 Methodology

The work will follow an iterative experimental software process. Each pipeline component will first be implemented as a measurable baseline, then improved only when error analysis demonstrates a specific weakness. The benchmark and evaluation protocol will be developed alongside the runtime system so that architecture decisions are based on observed performance rather than general model reputation.

6.1 Requirements and Threat Model

The team will confirm the document types, entity taxonomy, employee use cases, token scope, retention policy, target jurisdictions, hardware, and cloud task with Finaira. The initial threat model will protect raw sensitive text and reverse mappings from the external cloud provider and its request logs. It will assume that the local gateway and authorized employee environment remain trusted. Compromise of the local host, insider misuse, and universal protection against external background knowledge will be documented limitations rather than hidden assumptions.

6.2 Dataset and Annotation

The project will build a benchmark from controlled financial templates, public Arabic and multilingual resources, and human-reviewed Finaira-like examples that do not require real customer data. The test set will include Arabic-only, English-only, and code-switched text; Arabic-Indic and Western digits; transliteration; structured identifiers; hard negatives; ambiguous amounts and account numbers; repeated names; tables; and a limited OCR-noise slice. Entity type, sensitivity, expected policy action, and co-reference will be recorded separately.

Existing resources will be used as baselines and external checks rather than treated as a direct substitute for the project benchmark. TAB provides privacy-oriented annotations and co-reference in English legal text [10], while REDACT provides multilingual and code-switched PII slices

across a broad taxonomy [11]. Dataset splits will be separated by source or template family to reduce train-test leakage, and the final test set will be locked before threshold tuning.

6.3 Runtime Pipeline

Phase	Responsibility	Technical purpose
1	Document preparation	Extract text, tables, page structure, and provenance; invoke OCR only when needed.
2	Normalization	Create a detection view for Arabic variants, digits, spacing, and direction marks while keeping an offset map to the original.
3	Candidate detection	Run rules, checksums, PII models, general Arabic NER, and financial NER to maximize sensitive-span recall.
4	Fusion and context resolution	Merge overlaps, normalize labels, resolve aliases and ambiguous numbers, and use a local small model only for uncertain cases.
5	Privacy policy	Map each final entity to preserve, tokenize, generalize, remove, review, or block.
6	Tokenization and vault	Replace protected spans with scoped opaque tokens and encrypt the reverse mapping locally.
7	Pre-egress validation	Validate the fully assembled prompt, history, retrieved chunks, tokens, metadata, and unresolved risk before any cloud call.
8	Cloud processing	Run the selected task on sanitized text and require exact placeholder preservation.
9	Response validation	Reject invented, malformed, cross-session, or corrupted tokens and rescan for unexpected sensitive output.
10	Authorized restoration	Restore only valid tokens that the employee is authorized to view and record a privacy-safe audit event.

Table 1 Proposed Finaira runtime pipeline

For multi-turn employee conversations, the local system will maintain an encrypted raw transcript when retention is required, a sanitized conversation history for cloud context, and a local token vault. The cloud may receive a minimized sanitized summary and recent turns. Any selected case, customer, account, or document must be established by the authorized application state rather than guessed by a language model.

The system will treat pseudonymized data as protected personal data, not as anonymous data. The reverse mapping, original documents, keys, and access permissions will remain outside the cloud processing domain. This follows the separation principle described in pseudonymization guidance [12] and the risk-based evaluation approach recommended by NIST [13].

6.4 Proposed Technologies

Component	Candidate technology	Role and limitation
Document processing	PyMuPDF control baseline; Docling candidate	Docling supports structured conversion of PDFs and Office files [14].
OCR	PaddleOCR candidate	Arabic and English OCR support; used only if scanned documents are confirmed [15].
Structured detection	Python rules, checksum validators, libphonenumber, python-stdnum	Deterministic handling of known identifier formats.
Privacy and NER baselines	Presidio, AMWAL, Wojood-trained model, GLiNER2 PII	Complementary evidence; no single model is trusted as the safety boundary [5]-[7], [16].
Context resolver	Rules first; small local instruction model for uncertain spans	Avoids using a generative model on every span while supporting ambiguous context.
Policy and schemas	Versioned YAML or JSON with Pydantic	Deterministic and reviewable policy decisions.
Pseudonymization	Typed random or keyed tokens; authenticated encryption	Reversible mapping remains local and separate.
Storage	PostgreSQL for persistent mappings or Redis for short-lived sessions	Choice depends on confirmed retention and cross-document requirements.
Services	Python, FastAPI, Docker	Modular local gateway and reproducible deployment.
Retrieval	Local vector index and local raw document store	Only sanitized top-k context crosses the trust boundary.
Cloud integration	Provider adapter selected during implementation	Prevents the research from depending on one vendor.

Table 2 Proposed technology candidates subject to benchmarking

6.5 Privacy and Utility Evaluation

Evaluation area	Measures
Detection	Strict and overlap precision, recall, F1, recall-weighted F beta, and per-type or per-language results
Leakage	Documents containing any surviving high-risk raw entity, number and severity of leaked entities
Round trip	Exact reconstruction, token collisions, alias consistency, corrupted or invented token rejection
Retrieval	Recall at k, mean reciprocal rank, and normalized discounted cumulative gain against a raw local oracle
Question answering	Exact match or token F1 where applicable, evidence support, correctness, abstention, and restoration accuracy

Evaluation area	Measures
Contextual privacy	Bounded subject-inference experiment using sanitized text and defined external knowledge
Robustness	Arabic and English mixing, digit variants, OCR noise, invalid identifiers, aliases, and multiple subjects
System performance	Median and p95 latency, memory, throughput, cloud token use, and failure rate

Table 3 Proposed evaluation measures

6.6 Experiments

- Detector ablation comparing rules, individual models, the hybrid ensemble, and the ensemble with selective local contextual resolution.
- Protection-level comparison covering direct identifiers, selected quasi-identifiers, and more aggressive generalization.
- Token-scope comparison across request, document, and tenant or corpus scope to measure the utility gained against the linkability introduced.
- Retrieval comparison between a raw local oracle and an index built from sanitized text.
- Placeholder comparison between typed opaque tokens and realistic surrogate values as an optional extension.
- Robustness evaluation under code-switching, Arabic digit forms, altered spacing, OCR noise, and ambiguous financial numbers.

6.7 Safety and Ethical Constraints

No real customer data will be sent to an external service during development or evaluation. Logs will exclude raw documents, raw prompts, unprotected responses, and vault contents. The prototype will not claim regulatory certification or complete anonymity. Any employee authorization demonstrated by the prototype will be application-level and will not replace integration with a bank's production identity and access-management systems.

7.0 Expected Deliverables

The following are the expected outputs of the graduation project. They are prototype-level and research deliverables within the scope defined in Section 3.0, and do not constitute a production deployment or a regulatory certification.

7.1 Software Deliverables

- **Employee-facing privacy gateway prototype.** The integrated local gateway through which an authorized employee submits queries and documents, as shown in Figure 1.
- **Local document-processing and normalization pipeline.** Text extraction for plain text and text-layer PDF documents, with Arabic and English normalization and an offset map to the original text.

- **Hybrid sensitive-entity detection module.** Rules, checksum validators, Arabic and multilingual entity and PII models, and fusion with selective contextual resolution.
- **Policy engine for configurable privacy actions.** A versioned policy that assigns preserve, tokenize, generalize, remove, review, or block to each entity type.
- **Reversible pseudonymization and encrypted token vault.** Scoped opaque tokens with an encrypted local mapping and exact authorized restoration.
- **Pre-egress and post-response safety validation.** Gates that check residual PII, unresolved risk, token integrity, and unexpected sensitive output.
- **Local retrieval and cloud-based question-answering prototype.** Local raw document store and sanitized retrieval index, with only sanitized context sent to the cloud model through a provider adapter.
- **Logging and privacy-safe audit mechanism.** Audit events for restoration and operational logs that exclude raw documents, raw prompts, unprotected responses, and vault contents.

7.2 Research Deliverables

- **Human-reviewed Arabic and English financial privacy benchmark.** The annotated benchmark described in Section 6.2, with documented splits and annotation guidelines.
- **Experimental evaluation of detection approaches.** The detector ablation and baseline comparisons described in Section 6.6.
- **Privacy leakage analysis.** Document-level and entity-level leakage results, including the bounded subject-inference experiment.
- **Privacy-utility trade-off analysis.** Protection-level, token-scope, and retrieval comparisons relating privacy protection to retrieval and answer quality.
- **Robustness analysis.** Results under code-switching, digit variants, altered spacing, OCR noise, and ambiguous financial numbers.
- **Final technical documentation and research report.** Documentation of the design, the evaluation protocol, results, error analysis, and documented limitations.

8.0 Risks and Mitigation

Table 4 summarizes the principal technical and project risks, their expected impact, and the planned mitigations. The risks are consistent with the limitations and assumptions stated in Sections 3.0 and 6.1.

Risk	Impact	Mitigation
A sensitive entity is missed	High	Hybrid detection combining rules, validators, NER, and PII models with contextual resolution; pre-egress scanning of the fully assembled prompt; recall-weighted evaluation and false-negative thresholds confirmed with Finaira.
Excessive false positives reduce usefulness	High	Context resolution for ambiguous spans; configurable policy actions per entity type; hard negatives in the benchmark; privacy-utility comparison of protection levels.
The cloud model modifies, drops, or invents tokens	High	Exact placeholder preservation required in the cloud task; token integrity validation; invented, malformed, or cross-session tokens are rejected in response validation.
Arabic and English code-switching reduces detection quality	High	Dedicated mixed-language slice in the benchmark; robustness testing for digit variants, transliteration, and altered spacing; per-language reporting.
OCR introduces incorrect text	Medium	OCR is outside the core MVP and is invoked only if scanned documents are confirmed; confidence-based review and validation of OCR output where enabled; limited OCR-noise slice in the robustness evaluation.
The dataset does not represent realistic financial text	High	Controlled financial templates combined with human-reviewed Finaira-like examples and public resources; template-family splits; sponsor review of the taxonomy and examples.
Token mappings are exposed	Critical	Reverse mappings stored in a local encrypted vault with authenticated encryption; strict separation of vault, keys, and original documents from the cloud processing domain; logs exclude vault contents.
The project scope becomes too large	High	Explicit MVP boundaries with deferred features (Section 3.0); a single primary cloud task; iterative baselines improved only where error analysis shows a specific weakness.
Cloud provider behavior changes	Medium	Provider abstraction through an adapter so that the gateway does not depend on a single vendor; provider-independent evaluation of token preservation.
The local contextual model is too slow	Medium	Rules first, with the local small model invoked selectively for uncertain spans only; median and p95 latency measured in the system performance evaluation.

Risk	Impact	Mitigation
Contextual re-identification remains possible after span-level sanitization	High	Documented as a limitation rather than a hidden assumption; bounded subject-inference experiment; quasi-identifier handling and generalization compared across protection levels.
Requirements remain unconfirmed (formats, jurisdictions, hardware, retention, false-negative thresholds)	Medium	Confirm these choices with Finaira at the start of the project (Section 6.1); keep the recognizers, policy, and storage choices configurable.

Table 4 Risks and mitigation

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